Singapore School Kinderland

Continual Assessment 1

2026 - 2027

Secondary 2

Computer Science

Part 2 (Answer Key)

Date: _____ Name: **TEACHER ANSWER KEY**

Instructions:

- Start answering the questions on time.
- Follow all instructions carefully.
- Answer all questions.
- Submit your paper within the given time.

Total marks	**35**
Part 2	**30**
CW	**5**
Marks Obtained	**30 / 30**

Section A: Multiple Choice Questions (MCQ) (6x1=6 marks)

For each question 1-6, four options are given. Circle the correct option i, ii, iii or iv.

1. To convert a file size from bits to Bytes, what number do you divide by?

i. Divide by 2 **(ii.) Divide by 8**

iii. Divide by 1,000 **iv.** Divide by 1,024

2. How many Bytes are there in 1 Kibibyte (1 KiB)?

i. 8 Bytes **ii.** 100 Bytes

iii. 1,000 Bytes **(iv.) 1,024 Bytes**

3. How many Hertz (Hz) are equal to 1 Kilohertz (1 kHz)?

i. 8 Hz **ii.** 100 Hz

(iii.) 1,000 Hz **iv.** 1,024 Hz

4. In 8-bit ASCII text, how many Bytes are needed to store a word with 12 characters?

i. 6 Bytes **(ii.) 12 Bytes**

iii. 24 Bytes **iv.** 96 Bytes

5. Which formula calculates the file size (in bits) of a bitmap image?

i. Width + Height + Colour Depth **(ii.) Width × Height × Colour Depth**

iii. Width × Sample Rate **iv.** Height / 1,024

6. In digital audio recording, how many channels are used in a Stereo sound file?

i. 1 channel **(ii.) 2 channels**

iii. 4 channels **iv.** 8 channels

Section B: True or False (6x1=6 marks)

7. Write True or False in the box provided for each of the following statements:

No.	Statement	True / False
i.	Exactly 8 bits make up 1 Byte of computer memory.	True
ii.	1 Mebibyte (1 MiB) is equal to 1,024 KiB.	True
iii.	1 Megahertz (1 MHz) is equal to 1,000 kHz (or 1,000,000 Hz).	True
iv.	A Mono audio recording uses 2 separate sound channels.	False
v.	Spaces and punctuation marks in a text file also take up 1 Byte (8 bits) each in ASCII.	True
vi.	Increasing the colour depth of an image makes its file size smaller.	False

Section C: Fill in the Blanks (6x1=6 marks)

8. Fill in the blanks using the correct words from the box below:

Word Bank: [8 | 1,024 | 1,000 | Pixel | Stereo | Hertz (Hz)]

i. To convert a file size from bits to Bytes, we always divide the number of bits by **8**.

ii. To convert Bytes into Kibibytes (KiB), we divide the number of Bytes by **1,024**.

iii. A sample rate of 2 kHz is equal to 2 × **1,000** = 2,000 samples per second.

iv. The smallest single dot of colour in a bitmap image is called a **Pixel**.

v. A **Stereo** sound recording uses 2 audio channels (Left and Right).

vi. The unit used to measure sound sample rate (samples per second) is **Hertz (Hz)**.

Section D: Text & Image File Size Calculations (2x3=6 marks)

9. Calculate the file sizes below. Show your step-by-step working clearly:

i. (3m)	**Text File Size:** A plain text story has **2,048 characters** in 8-bit ASCII (1 Byte per character). Find size in **(a) bits**, **(b) Bytes**, and **(c) KiB**: **(a) Bits = 2,048 × 8 = 16,384 bits | (b) Bytes = 2,048 Bytes | (c) KiB = 2,048 / 1,024 = 2 KiB**
ii. (3m)	**Image File Size:** A bitmap icon is **64 × 32 pixels** with an **8-bit colour depth**. Find size in **(a) bits**, **(b) Bytes**, and **(c) KiB**: **(a) Bits = 64 × 32 × 8 = 16,384 bits | (b) Bytes = 16,384 / 8 = 2,048 Bytes | (c) KiB = 2,048 / 1,024 = 2 KiB**

Section E: Sound File Size Calculations (2x3=6 marks)

10. Use (Sample Rate × Resolution × Time × Channels) to calculate the audio file sizes:

i. (3m)	**Mono Sound File:** Sample Rate = **1,000 Hz**, Resolution = **8 bits**, Time = **8 seconds**, **Mono (1 channel)**. Find size in **(a) bits** and **(b) Bytes**: **(a) Bits = 1,000 × 8 × 8 × 1 = 64,000 bits (2m) | (b) Bytes = 64,000 / 8 = 8,000 Bytes (1m)**
ii. (3m)	**Stereo Sound File:** Sample Rate = **1,024 Hz**, Resolution = **16 bits**, Time = **4 seconds**, **Stereo (2 channels)**. Find size in **(a) Bytes** and **(b) KiB**: **(a) Bits = 1,024 × 16 × 4 × 2 = 131,072 bits → Bytes = 16,384 Bytes (2m) | (b) KiB = 16,384 / 1,024 = 16 KiB (1m)**